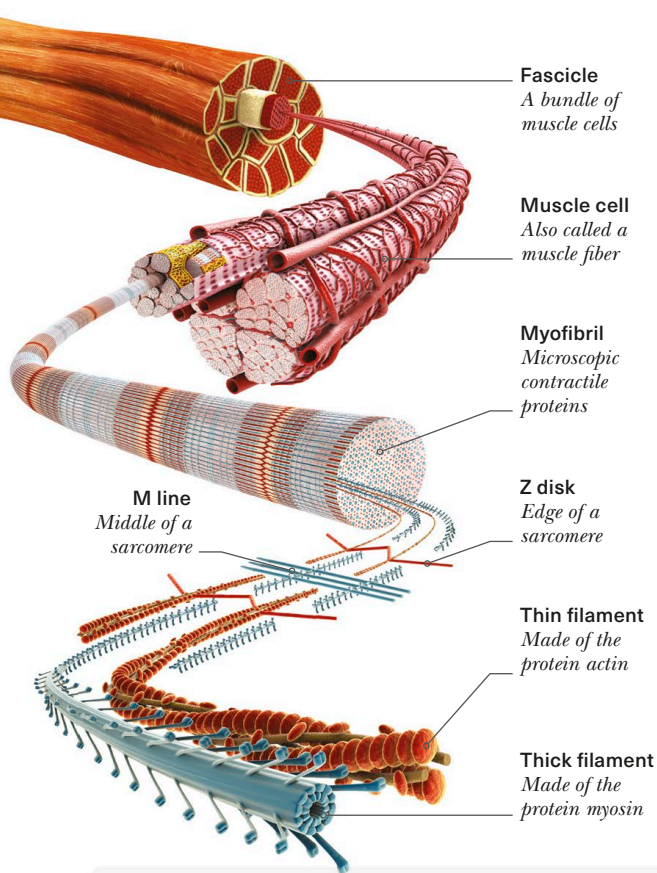


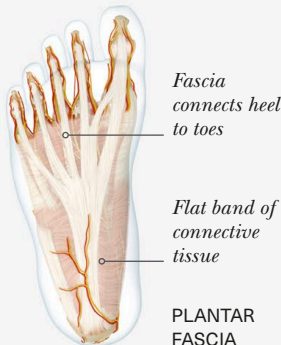
MUSCLE STRUCTURE

Skeletal muscles are bundles of bundles of bundles of parallel muscle cells, blood vessels, and nerves wrapped with connective tissue, including fascia. Fascia creates a network through and around muscles and other structures of your body. Microscopic proteins in your muscles cause muscle contractions.



Fascia

Fascia is similar to the white pith of an orange; it separates parts yet integrates the whole. Fascia is not just found around muscles. It is also around vital organs and woven throughout your body. This body-wide network of fascia is part of the reason why a yoga pose that affects your foot can suddenly release your tight shoulders.

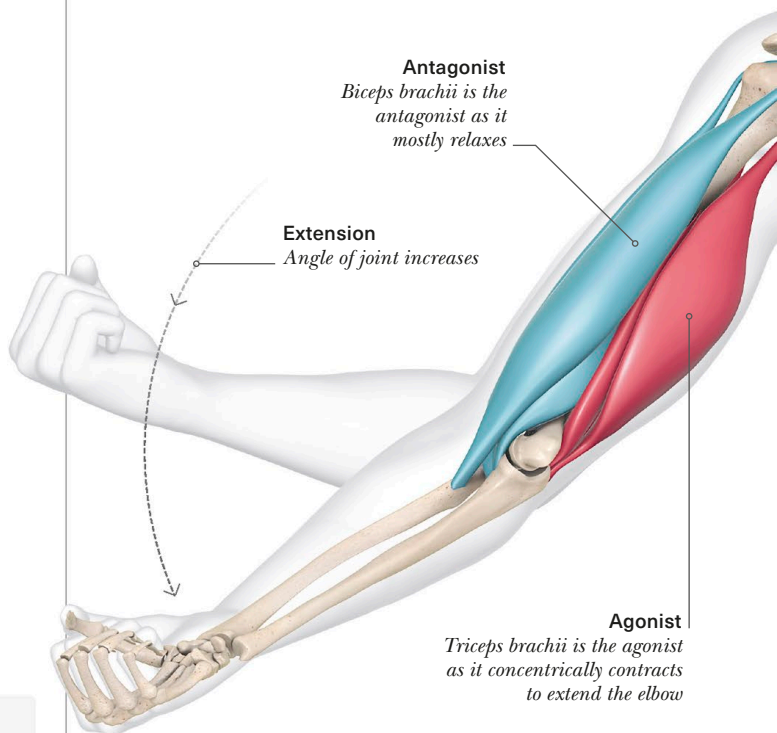


HOW MUSCLES WORK

Muscles often work in antagonistic pairs. As the agonist muscle engages, the antagonist generally releases. Synergist muscles engage around the joint to support the action.

TYPES OF CONTRACTION

Isotonic contractions involve a change in muscle length, as in the act of flexing or extending your elbow (see below) or transitioning in or out of an asana. Isometric contractions involve tension with no change in muscle length, such as when holding an asana.



ECCENTRIC CONTRACTION

Eccentric contractions occur when muscle fibers “lengthen” to change the angle of a joint. This occurs in your biceps when extending your elbow as you lower a weight, or in your hamstrings when extending your knee as you transition from Warrior II to Triangle (see pp.118–21) pose.